From Linear Scores to a Single Hidden Layer

A Mathematical Study of Simple Learning Systems

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The Central Question

Research Question

When does a one-hidden-layer nonlinear classifier genuinely improve on a linear decision rule, and when is additional model complexity unnecessary?

Our approach - argue from evidence, not preference:

- Compare **softmax regression** (linear baseline) against a **one-hidden-layer neural network** ($\tanh$ + softmax)
- Three datasets of increasing geometric complexity: Gaussian blobs, moons, digits
- Derive the mathematics that each model relies on, then test experimentally
- **More complexity is not automatically better**, if the linear baseline is sufficient, we say so and justify it

Why the Comparison is Fair

Datasets:

- **Linear Gaussian** (n=400), **Moons** (n=400)
- **Digits** (n=1,797, d=64, K=10)

Shared across models:

- **Train/Val/Test Split:** 60/20/20
- Preprocessing: Normalization only
- L_2 regularisation: $\lambda = 10^{-4}$
- Epoch budget: 200
- Mini-batch size: 64

Model selection protocol:

- Best checkpoint chosen on **validation CE only**
- Test set touched **exactly once** per seed
- 5 random seeds with 95% CI reporting

Two Models, Same Objective

Model 1: Softmax Regression

$$s(x) = Wx + b$$

$$p_j(x) = \frac{\exp(s_j)}{\sum_\ell \exp(s_\ell)}$$

- **Geometry:** hyperplane boundaries
- 650 params on Digits ($Kd + K$)
- Represents exactly the family of affine classifiers

Model 2: One-Hidden-Layer NN

$$h = \tanh(W_1 x + b_1)$$

$$s = W_2 h + b_2$$

- **Geometry:** curved boundaries
- 2,410 params on Digits ($3.7\times$ more)
- Hidden layer = learned nonlinear coordinate change

Objective: Negative Log-Likelihood

Before optimizing, we define our loss function. Both models minimize the **Cross-Entropy (CE) Loss**, which is mathematically equivalent to the Negative Log-Likelihood for a categorical distribution:

$$\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{y_i}(x_i)$$

- **Probabilistic meaning**: Maximizes the likelihood of the training data.
- **Optimization**: Provides clean, numerically stable gradients when paired with softmax.

Backpropagation: Chain Rule in Action

Idea: propagate loss sensitivities backward through the computation graph.

Key gradient formulas (vectorized):

$$\frac{\partial \mathcal{L}}{\partial S} = \tfrac{1}{n}(P - Y)$$

$$\frac{\partial \mathcal{L}}{\partial W_2} = \left(\tfrac{\partial \mathcal{L}}{\partial S}\right)^\top H$$

$$\frac{\partial \mathcal{L}}{\partial Z_1} = \tfrac{\partial \mathcal{L}}{\partial S} W_2 \odot (1 - H \odot H)$$

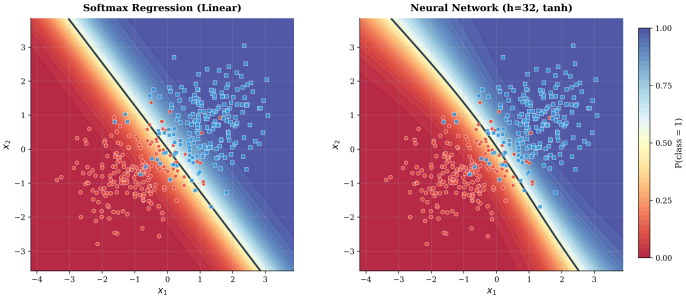
$$\frac{\partial \mathcal{L}}{\partial W_1} = \left(\tfrac{\partial \mathcal{L}}{\partial Z_1}\right)^\top X$$

Implementation Sanity Checks

Before running experiments, we rigorously verified the from-scratch implementation:

Check	Result	Status
Probability Normalization	Softmax rows sum exactly to 1.0	Pass
Monotonic Loss Decrease	Loss strictly decreases ($1.62 \to 1.34$)	Pass
Numerical Gradient Check	Max error $< 10^{-11}$	Pass
No NaN/Inf Detection	Parameters remain finite after training	Pass
Micro-dataset Overfit	10 samples fit to 100% training accuracy	Pass

Experiment 1: Linear Gaussian — Linear Model is Sufficient



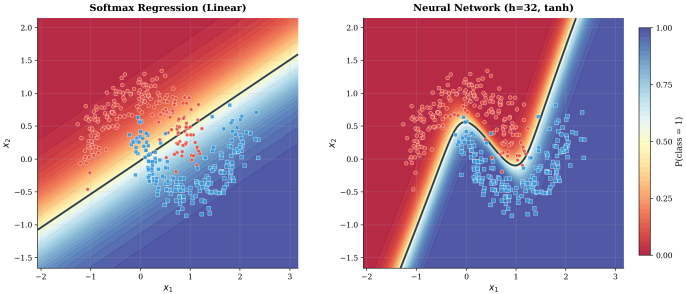
Key Finding

Bayes-optimal boundary is a hyperplane $\Rightarrow$ **linear model suffices**.

- Both models learn the same boundary
- NN's extra parameters are wasted

Takeaway: Complexity does not justify itself here

Experiment 2: Moons — Nonlinearity is Essential



Key Finding

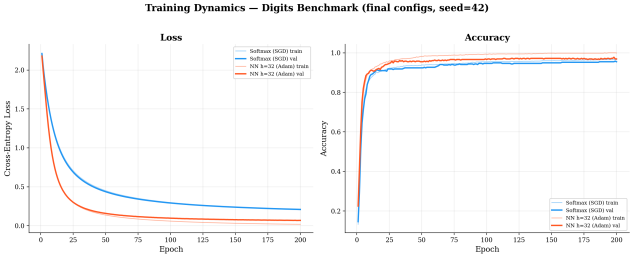
Interleaving arcs are **not linearly separable** $\Rightarrow$ no hyperplane can separate them.

- Softmax places a line through both classes $\Rightarrow$ fails
- NN learns a curved boundary $\Rightarrow$ near-100% accuracy

What the hidden layer does:

$h = \tanh(W_1 x + b_1)$ maps $\mathbb{R}^2 \to \mathbb{R}^{32}$. In the new space, the arcs become linearly separable; the output layer draws a hyperplane there.

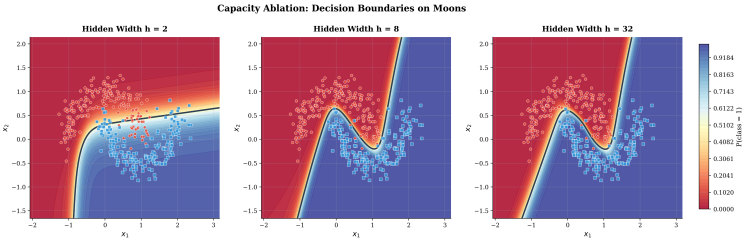
Experiment 3: Digits Benchmark — Real-World Task



Model	Acc	CE
Softmax	93.3%	0.269
NN $h = 32$	**95.7%**	**0.157**

- $\sim$2.5 ppt accuracy gain, 42% lower CE
- CE reduction $>$ accuracy gain $\Rightarrow$ NN gives better-calibrated probabilities
- Linear model is already strong (93%) Digits is “mostly linear”

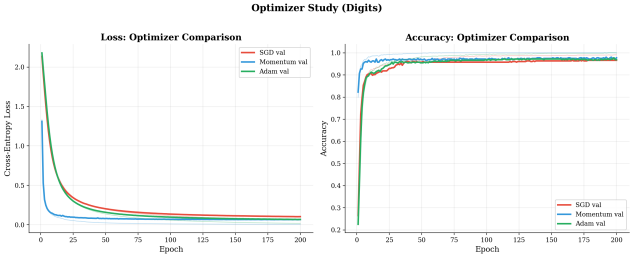
Capacity Ablation: Hidden Width on Moons $\{2, 8, 32\}$



- $h = 2$: crude, nearly linear separation - too few dimensions to represent the curve
- $h = 8$: smooth nonlinear boundary captures the moon shape
- $h = 32$: no significant difference from $h = 8$

Takeaway: capacity must match task complexity; beyond that, returns diminish.

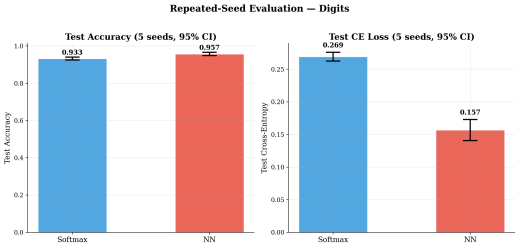
Optimizer Study on Digits (NN $h = 32$, fixed hyperparams)



Optimizer	Acc	CE	Ep.
SGD	94.8%	0.170	197
Momentum	**95.4%**	0.165	197
Adam	95.1%	**0.151**	197

- All three reach similar final accuracy ($<$0.6 ppt spread)
- Adam: lowest CE (best calibration), fastest early convergence
- Momentum: highest accuracy
- **Optimizer choice is secondary to model-class choice** ($\sim$0.5 ppt vs. $\sim$2.5 ppt)

Repeated-Seed Evaluation (5 Seeds, 95% CI)



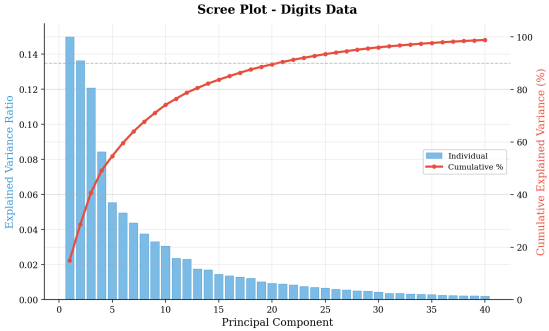
$$\bar{x} \pm t_{0.025,\,4} \cdot \frac{s}{\sqrt{5}} = \bar{x} \pm 2.776 \cdot \frac{s}{\sqrt{5}}$$

	Test Accuracy	Test CE
Softmax	$93.3\% \pm 0.7\%$	0.269 ± 0.007
NN	$\mathbf{95.7\% \pm 0.9\%}$	$\mathbf{0.157 \pm 0.016}$

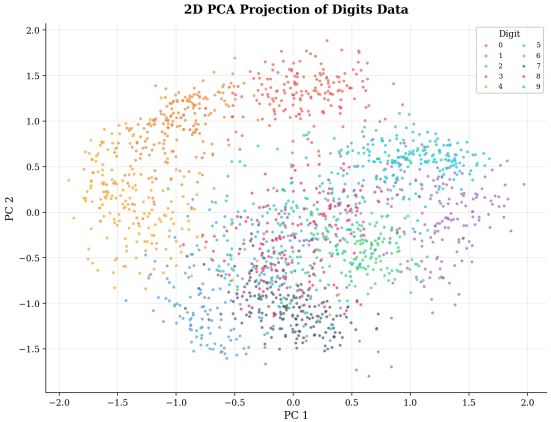
What repeated seeds add beyond a single run:

- CIs do not overlap $\Rightarrow$ NN advantage is **robust**, not a lucky seed
- NN has wider CI ($\pm$0.9% vs. $\pm$0.7%) — more params $\Rightarrow$ more init sensitivity
- Without this, we cannot distinguish genuine model-class gains from random variation

Track A: PCA/SVD — How Much Linear Structure Exists?



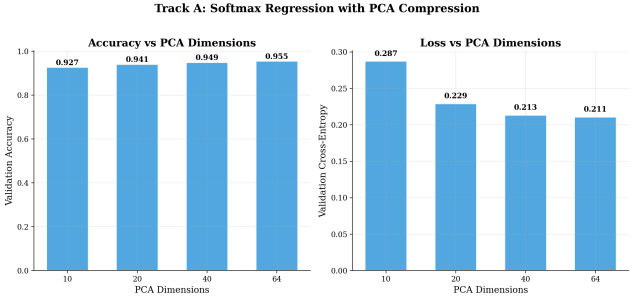
Rapid decay: top 20 PCs capture **89.6%** of variance



2D: digits 0,6 separate; 3,5,8 overlap (visual similarity)

Key insight: The pixel space is effectively ∼20-dimensional - most variation lives in a low-rank subspace.

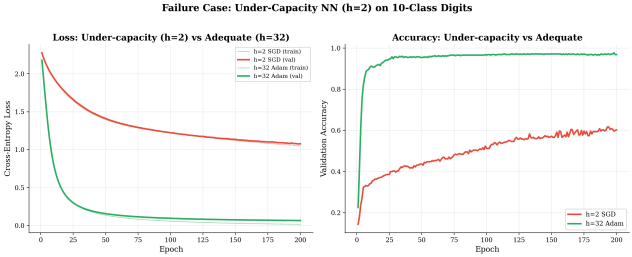
Track A: Softmax Performance vs. PCA Dimensions



PCA m	Val Acc	Val CE
10	92.7%	0.287
20	94.1%	0.229
40	94.9%	0.213
64 (full)	95.5%	0.211

- $m{=}20$: discard 69% of dims, lose only 1.4 ppt
- Digits is “mostly linear” - linear methods already capture most discriminative structure
- NN at full 64 dims (95.7%) still beats Softmax at any PCA level $\Rightarrow$ **nonlinear features add value beyond what PCA captures**

Failure Case: Under-Capacity NN ($h=2$) on 10-Class Digits



	Acc	CE
Softmax (linear)	93.3%	0.269
NN $h=2$	59.0%	1.113
NN $h=32$	95.7%	0.157

Why it fails: capacity, not optimisation:

- 2-D bottleneck **destroys** 64-D information
- Same optimizer & epochs as $h=32 \Rightarrow$ not optimizer
- Train + val loss stay high $\Rightarrow$ **underfitting**
- **Lesson:** Capacity must match task complexity before optimisation can help.

Answering the Core Questions

- **Q1: When is the linear model geometrically sufficient?**
  When the true data boundary is flat (Gaussian), but it fails when boundaries must curve (Moons).
- **Q2: What representational change does the hidden layer provide?**
  It learns a nonlinear transformation that “untwists” complex data into a linearly separable space.
- **Q3: When does additional complexity fail to justify itself?**
  When the simple baseline already captures the geometry, or if the capacity is severely bottlenecked.
- **Q4: What does the failure case teach us?**
  It demonstrates that representational capacity is a strict prerequisite that optimisation cannot substitute.
- **Q5: What do repeated-seed statistics add beyond a single run?**
  They prove that the neural network’s advantage is robust across random initialisations, not just a lucky artefact.

Limitations — What Our Study Does Not Show

- **Architecture constraint**: Single hidden layer, $\tanh$ only — deeper networks or ReLU may behave differently.
- **Data scale**: Small-scale data ($\sim 1{,}800$ samples, $d{=}64$) — conclusions may not transfer to ImageNet-scale tasks.
- **Regularisation**: Only L_2 regularisation ($\lambda{=}10^{-4}$); no dropout, batch norm, or cross-validation.
- **Statistical limits**: 5 seeds assume approximate normality; optimizers were tested at one fixed setting.

Our findings reliably characterise this specific comparison on these three tasks under this fixed protocol.

Summary of Key Findings

Task	Linear model	Hidden layer
Gaussian	Sufficient; Bayes boundary is a hyperplane	No benefit; wasted parameters
Moons	Fails; cannot separate curved arcs	Essential; learns nonlinear coordinate change
Digits	Strong (93.3%); mostly linear structure	Modest gain (95.7%); captures residual nonlinear features
Digits $h=2$	93.3%; better!	59.0%; bottleneck destroys information

Thank You

Questions?